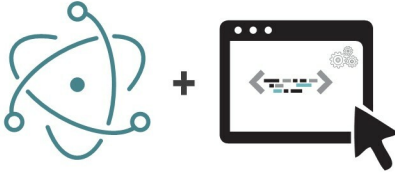


Electron: Building Cross-platform Desktop Apps with Web Technologies

The programming languages used to build desktop and Web applications are obviously different. Electron is an effort to bridge this gap. With it, you can construct a desktop application with popular Web technologies such as HTML, CSS and JavaScript. This article introduces the basic features and the potential of the Electron framework.



There are three major types of software applications that are built — desktop, Web and mobile. The programming platforms used to construct each of these types of apps are different. Though certain languages such as Java play a key role in each of these application types, there are specialised languages for each category. To name a few, PHP is for server-side scripting, CSS for adding style to Web pages, and XML is for data representation. Similarly, there are frameworks specific to each of these categories. These make the application-building process easier and more efficient for developers.

Besides all these options, there are also a few frameworks that developers can use to build both desktop and mobile applications. This article introduces you to one such popular framework named Electron. The three major characteristics of the Electron framework are listed below:

- It uses popular Web technologies such as HTML, CSS and JavaScript for building desktop applications.
- It is open source.
- It supports cross-platform application development. It is compatible with Microsoft Windows (Windows 7 and above), Apple Mac and Linux. The applications built using Electron execute on all these operating systems.

Building apps using the Electron framework is fairly simple. Primarily, Electron was built for the Atom editor (Atom is a popular text editor/IDE — <https://atom.io/>). The efficiency of Electron is proven by the many leading companies such as Facebook and Microsoft that use it. A sample list of products built using Electron is available in the official home page (<https://electron.atom.io/>).

Installing Electron

Electron can be installed easily by using the `npm install`

command:

```
npm install electron --save-dev --save-exact
```

Obviously, you need to have Node.js installed on your computer.

Quick starting Electron

The quickest way to get started with Electron is to get the 'Quick start' app into action. For that you have to carry out the following steps:

```
# Clone the Quick Start repository using Git
$ git clone https://github.com/electron/electron-quick-start

# Navigate to the folder electron-quick-start
$ cd electron-quick-start

# Install the dependencies and run
$ npm install && npm start
```

Another easier approach to get familiar with Electron is to check out the Electron-API-Demos from GitHub. The steps to build this demo app are the same as those mentioned for the 'Quick start' app.

```
$ git clone https://github.com/electron/electron-api-demos

$ cd electron-api-demos
$ npm install
$ npm start
```

This app can be used to get a basic introduction to what can be done with Electron. You can play around with